Lecture 5, Dec 23, 2025.

Thm. The following types of functions are continuous at every number in their domains.

- Polynomials ($\mathrm{Dom} = \mathbb{R}$).
- Rational functions ($\mathrm{Dom}\left(\frac{f}{g}\right) = (\mathrm{Dom}\, f \cap \mathrm{Dom}\, g) \setminus \{g = 0\}$).
- root functions $\sqrt[n]{f(x)}$
  - n is even: $\mathrm{Dom}\sqrt[n]{f} = \mathrm{Dom}\, f \cap \{f(x) \geq 0\}$
  - n is odd: $\mathrm{Dom}\sqrt[n]{f} = \mathrm{Dom}\, f$.
- trigonometric functions, $\mathrm{Dom}\sin x = \mathbb{R}$, $\mathrm{Dom}\cos x = \mathbb{R}$.

Example. On what intervals is each function continuous?

a). $f(x) = x^{100} - 2x^{37} + 75$.

Sln. f is a polynomial, so it is continuous on $(-\infty, \infty)$.

b). $g(x) = \dfrac{x^2 + 2x + 17}{x^2 - 1}$

Sln. Notice that $\mathrm{Dom}\, g = \mathbb{R} - \{1, -1\}$ then since g is a fraction g is continuous on $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.

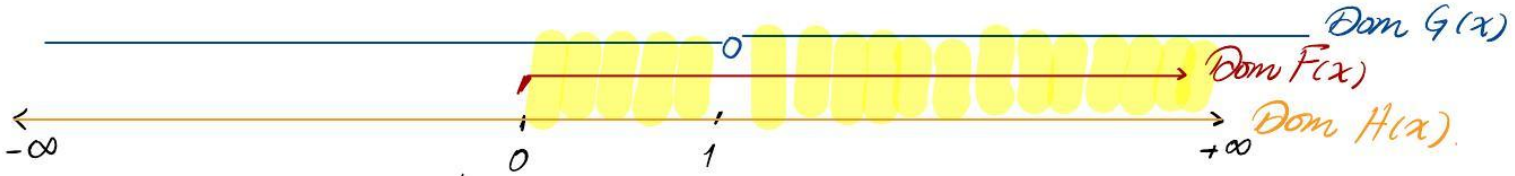
c). $h(x) = \sqrt{x} + \dfrac{x+1}{x-1} - \dfrac{x+1}{x^2+1}$

Sln. We can write $h(x) = F(x) + G(x) - H(x)$ where

$$F(x) = \sqrt{x}, \quad G(x) = \frac{x+1}{x-1}, \quad H(x) = \frac{x+1}{x^2+1}$$

$\mathrm{Dom}\, F(x) = [0, \infty)$ $\mathrm{Dom}\, G(x) = \mathbb{R} - \{1\} = (-\infty, 1) \cup (1, \infty)$

$\mathrm{Dom}\, H(x) = (-\infty, \infty)$



$\mathrm{Dom}\, h(x) = [0, 1) \cup (1, \infty)$

Example. Evaluate $\lim_{x \to \pi} \frac{\sin x}{2 + \cos x}$.

Sln. By the thm $y = \sin x$ is continuous on $\mathbb{R}$ and $y = 2 + \cos x$ is also continuous on $\mathbb{R}$. Now, $\frac{\sin x}{2 + \cos x}$ is continuous for x s.t $2 + \cos x \neq 0$ but notice that

$$\cos x \geq -1. \quad \text{then}$$

$$2 + \cos x \geq -1 + 2 = 1$$

So, $2 + \cos x \neq 0$ for all $x \in \mathbb{R}$ then,

$$f(x) = \frac{\sin x}{2 + \cos x} \quad \text{is continuous on } \mathbb{R}.$$

in particular $f(x)$ is continuous at $x = \pi$.

$$\lim_{x \to \pi} \frac{\sin x}{2 + \cos x} = f(\pi) = \frac{\sin \pi}{2 + \cos \pi} = 0.$$

Thm. If f is a continuous at b and $\lim_{x \to a} g(x) = b$ then

$$\lim_{x \to a} f(g(x)) = f(b).$$

in other words

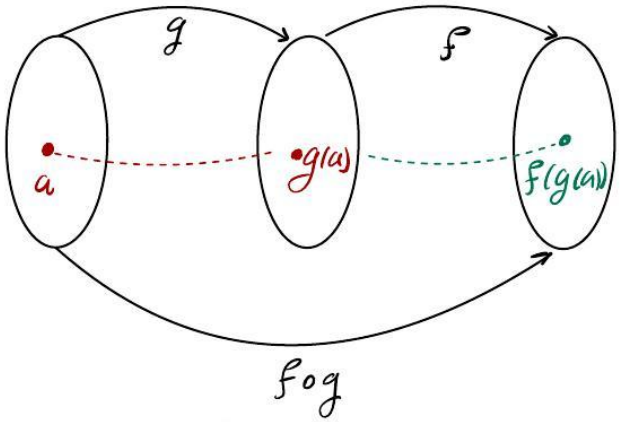
$$\lim_{x \to a} f(g(x)) = f\left(\lim_{x \to a} g(x)\right).$$

For example, $f(g(x)) = \sqrt[n]{g(x)}$ and.

$$\lim_{x \to a} f(g(x)) = \lim_{x \to a} \sqrt[n]{g(x)}$$

$$= \sqrt[n]{\lim_{x \to a} g(x)}.$$

Thm. If g is continuous function at a and f is continuous at $g(a)$ then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a.



Example. where are the following functions continuous?

a). $h(x) = \sin(x^2)$ b). $F(x) = \dfrac{1}{\sqrt{x^2+7} - 4}$

Sln. a) Notice that

$$h(x) = f(g(x)) \quad \text{where}$$

$$f(x) = \sin x, \quad g(x) = x^2$$

$$(f(g(x)) = f(x^2) = \sin x^2 = h(x))$$

Now, g is continuous on $\mathbb{R}$ (it's a polynomial) f is also continuous everywhere. Then by thm $h = fog$ is continuous everywhere.

b). $F(x) = f(g(x))$ where $f(x) = \dfrac{1}{\sqrt{x} - 4}$ $g(x) = x^2 + 7$.

Notice that g is a continuous function on $\mathbb{R}$ b/c it's a polynomial.

for $f(x)$ since it's a fraction Solving $\sqrt{x} - 4 = 0$.

$$\sqrt{x} = 4$$

$$(\sqrt{x})^2 = 4^2$$

$$x = 16.$$

then, $\text{Dom } f = \mathbb{R} - \{16\}$ for $F(x)$ we need compute.

$$\sqrt{x^2+7} - 4 = 0$$

$$\sqrt{x^2+7} = 4$$

$$(\sqrt{x^2+7})^2 = 16$$

$$x^2 + 7 = 16$$

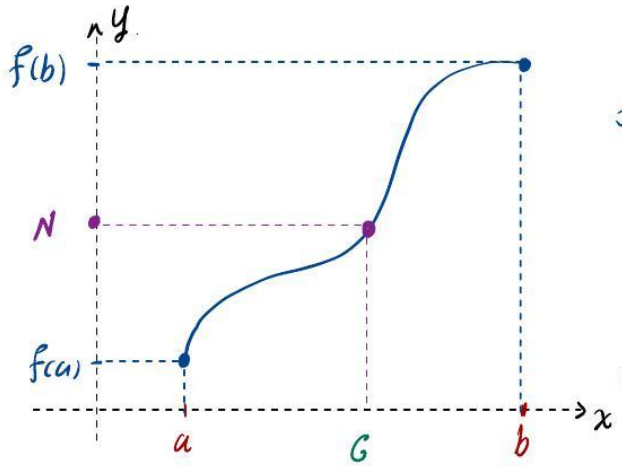
$$x^2 = 9$$

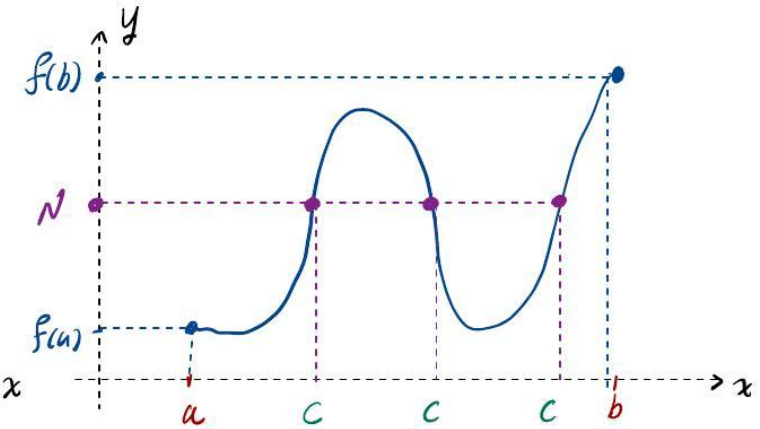
$$x = \pm 3.$$

$\text{Dom } F(x) = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)$.

The intermediate Value Theorem.

Thm. Suppose that f is continuous on the closed interval $[a, b]$ and N be a number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then, there exists a number $c \in (a, b)$ such that $f(c) = N$.





Example. Show that there is a solution of the equation.

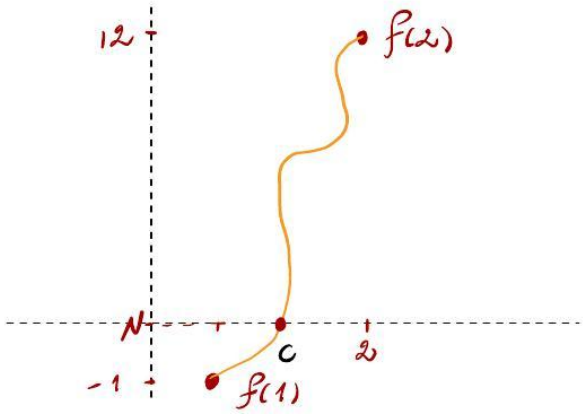
$$4x^3 - 6x^2 + 3x - 2 = 0 \quad (*)$$

between 1 and 2.

Sln: Let $f(x) = 4x^3 - 6x^2 + 3x - 2$ $(*)$ iff $f(x) = 0$

Notice that $f(x)$ is continuous on $[1, 2]$. (b/c f is a polynomial then is continuous on $\mathbb{R}$ in particular it's continuous on $[1, 2]$).

Then, $f(1) = 4 - 6 + 3 - 2 = -1 < 0$ $\quad f(2) = 4(8) - 6(4) + 6 - 2 = 12 > 0$.



So, if $N = 0$, $f(1) < 0 = N < f(2)$

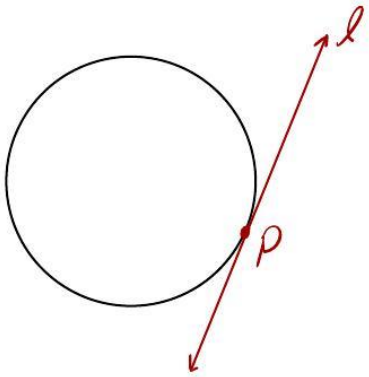
it's a number between $f(1)$ and $f(2)$

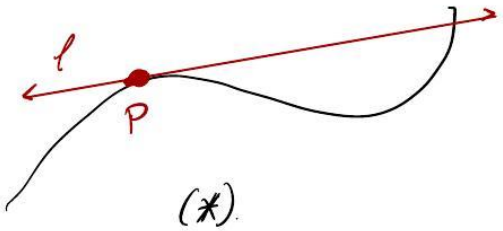
then since f is continuous by intemediate Value theorem there is a number c between $(1, 2)$ s.t $f(c) = 0$

that $4c^3 - 6c^2 + 3c - 2 = 0$ which is the solution for the equation (III) between 1 and 2.

The tangent and Velocity Problems

The tangent problem. The word tangent means "touching". We can think of a tangent to a curve as a line that touches the curve and follows the same direction as the curve at the point of contact.





(*)

- For the circle we can say that the tangent line is a line l s.t intersects the circle once and only once.
- For more general curves this definition is not correct in (*) l is tangent at P but intersects the curve.

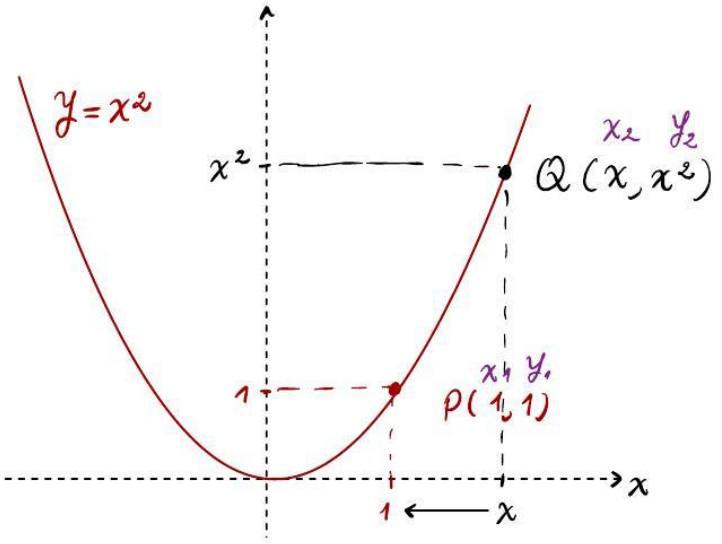
Example. Find an equation of the tangent line to the curve $y = x^2$ at the point $(1, 1)$.

Sln. For a line we require two elements.

i). slope and a Point

ii). two points

For compute the slope we need at least two points. Here we have a proplm b/c we have just one point.



then we can compute the slope of the line $\overleftrightarrow{PQ}$.

$$m_{\overleftrightarrow{PQ}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{x^2 - 1}{x - 1} \qquad P \neq Q \quad x \neq 1$$

For example: $x = 2$.

$$m_{\overleftrightarrow{PQ}} = \frac{4 - 1}{2 - 1} = 3.$$ but the line is not tangent at P.

If we move the point Q to P then we have a tangent line.

If $Q \to P$ then that means $x \to 1$. So, we say that the slope of the tangent line is the limit of the slope of the secant lines.

$$\lim_{Q \to P} m_{\overleftrightarrow{PQ}} = m \quad \text{and} \quad \lim_{x \to 1} \frac{x^2 - 1}{x - 1} = 2.$$

Now, using the point-slope form of the equation m - slope, $P - (x_1, y_1)$

$$y - y_1 = m(x - x_1)$$

Here, $x_1 = 1$, $y_1 = 1$ and $m = 2$.

$$y - 1 = 2(x - 1)$$

$$y = 2x - 2 + 1$$

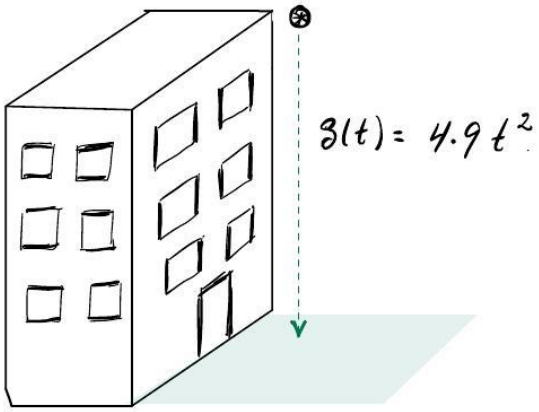
$$y = 2x - 1 \qquad (y = mx + b).$$

The Velocity Problem. Find the instantaneous velocity of an object moving along a straight path at a specific time if the position of the object at any time is know. In the next example, we investigate the velocity of a falling ball.

In the distance fallen after t seconds is denoted by $s(t)$ and measured in meters then the Galileo's observation says that

$$s(t) = 4.9t^2.$$

Example. Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground. Find the velocity of the ball after 5 seconds.

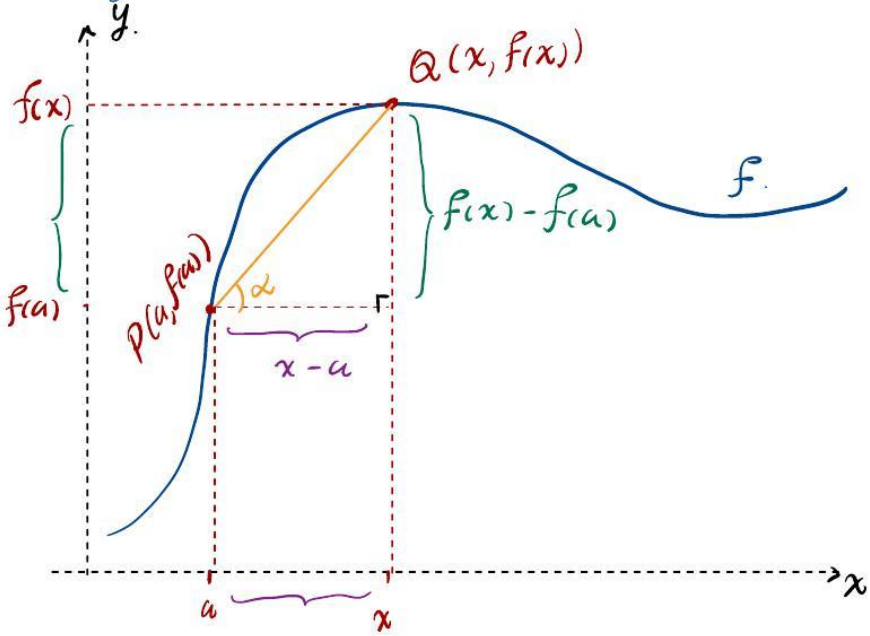


Sln. The problem here is finding the instantaneous velocity at $t=5$.

$$\text{average Velocity} = \frac{\text{change in position}}{\text{change in time.}}$$

Derivatives and Rates of change.

* tangents



$$m_{PQ} = \tan\alpha = \frac{f(x) - f(a)}{x - a}$$

Def. The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$ is the line through P with slope

$$m = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}.$$

Provided that this limit exists.

$$v\big|_{t=5} = \lim_{t\to 5} \text{average velocity} = \lim_{t\to 5} \frac{s(t)-s(5)}{t-5}$$

recall that $s(t) = 4.9t^2$ then $s(5) = 4.9(5)^2$

$$v\big|_{t=5} = \lim_{t\to 5} \frac{4.9t^2 - 4.9(5)^2}{t-5}$$

$$= \lim_{t\to 5} \frac{4.9(t^2-5^2)}{t-5}$$

$$= 4.9 \lim_{t\to 5} \frac{(t-5)(t+5)}{t-5}$$

$$= 4.9 \lim_{t\to 5} (t+5) = 4.9(10) = 49 \ m/s.$$

Rmk. we know that $m = \lim_{x\to a} \frac{f(x)-f(a)}{x-a}$, put $h = x-a$, $x = h+a$

$$m = \lim_{h\to 0} \frac{f(h+a)-f(a)}{h}$$

Example Find an equation of the tangent line the hyperbola $y = 3/x$ at the point $(3,1)$, $a \ \ f(a)$

Sln:

$$m = \lim_{h\to 0} \frac{f(h+3)-f(3)}{h}$$

$$= \lim_{h\to 0} \frac{\frac{3}{h+3} - \frac{1}{1}}{h}$$

$$= \lim_{h\to 0} \frac{\frac{3-(h+3)}{h+3}}{\frac{h}{1}}$$

$$= \lim_{h\to 0} \frac{3-h-3}{h(h+3)}$$

$$= \lim_{h\to 0} \frac{-h}{h(h+3)}$$

$$= -\lim_{h\to 0} \frac{1}{h+3} = -\frac{1}{3}.$$

x_1 y_1

$P(3,1)$

then $y - y_1 = m(x - x_1)$

$$y - 1 = -\frac{1}{3}(x-3)$$

$$y = -\frac{1}{3}x + 1 + 1$$

$$y = -\frac{1}{3}x + 2$$

Derivatives

Def. The derivative of a function f at a number a denoted by $f'(a)$ is

$$f'(a) = \lim_{h\to 0} \frac{f(h+a) - f(a)}{h}$$

$$f'(a) = \lim_{x\to a} \frac{f(x) - f(a)}{x - a}$$

if this limit exists.

Example. Use the definion of derivative to find $f'(2)$ if $f(x) = x^2 - 8x + 9$.

Sln. $$f'(2) = \lim_{h\to 0} \frac{f(h+2) - f(2)}{h}$$

$f(h+2) = (h+2)^2 - 8(h+2) + 9 = h^2 + 4h + 4 - 8h - 16 + 9 = h^2 - 4h - 3.$

$f(2) = 2^2 - 8(2) + 9 = 4 - 16 + 9 = -3.$

$$f'(2) = \lim_{h\to 0} \frac{[h^2 - 4h - 3] - [-3]}{h}$$

$$= \lim_{h\to 0} \frac{h^2 - 4h - 3 + 3}{h}$$

$$= \lim_{h \to 0} \frac{h^2 - 4h}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{h}(h-4)}{\cancel{h}}$$

$$= \lim_{h \to 0} (h-4) = -4.$$

For you! Compute $f'(a)$ and $f'(5)$, $f'(10)$.